

Sreejato (Sree) Chatterjee

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EDUCATION

University of Rochester

Rochester, NY

BS in Computational Biology, BA in Data Science

Aug 2022 – May 2026

- **Relevant Coursework:** Computational Biology w/Lab, Applied Genomics w/Lab, Applied Population Biology w/Lab, Data Mining, Artificial Intelligence, Data Structures, Intermediate Statistical Methodology, Tools for Data Science, Genetics w/Lab, Biochemistry, Human Physiology, Linear Algebra with Differential Equations, Statistical Machine Learning, Machine Learning for Limited Data

RELEVANT SKILLS

- **Programming Languages:** Java, Python, R, MATLAB, Bash, SAS, HTML, PHP, SQL
- **Technologies:** ImageJ, DeepLabCut, Git, Linux
- **Technical Frameworks:** NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, Seaborn, PyTorch, TensorFlow, Spark
- **Lab Techniques:** PCR, qPCR, Gel Electrophoresis, Western Blotting, Cell Culture, Flow Cytometry, DNA Extraction, Immunohistochemistry, Tissue Sectioning, ELISA, Fluorescence Microscopy

PUBLICATIONS

1. **Chatterjee, S.,** Tran, L., Nguyen, Q. D., Kirson, R., Hamlin, D., Aquino, H., Lyu, H., Luo, J., & Dye, T. (2025). *Assessing Historical Structural Oppression Worldwide via Rule-Guided Prompting of Large Language Models*. In Proceedings of the 2025 IEEE International Conference on Big Data (11th Special Session on Intelligent Data Mining). arXiv:2509.15216.

HONORS & AWARDS

- First Place in Biomedical Data Science Hackathon Fall 2024 & 2025 (Undergraduate Division)
- Neurodiversity Alliance Scholarship Recipient 2025 (\$2,500/ year)
- 2nd Place Undergraduate Division, RAISE AI, Informatics & Data Science Competition 2025
- 1st Place Gold Award, Visualization Category, RIT ASA Datafest 2025
- Out 4 Undergrad Life Sciences Conference Scholar 2024
- Dean's Scholarship Recipient (\$20,000/year) 2022-2026

RESEARCH EXPERIENCE

University of Rochester School of Medicine & Dentistry (SMD)

Rochester, NY

Computational Genomics Researcher, Department of Biomedical Genetics (PI: Dr. Hongbo Liu)

Jun 2025 – Present

- Conducting large-scale genomic analyses to identify genome-wide significant variants across thousands of GWAS traits and quantify their enrichment within transposable elements (LINEs, SINEs, LTRs, etc.) and enhancer-like regulatory regions, with a focus on brain aging & neurogenetics.
- Designing and implementing bioinformatics pipelines for large-scale variant processing and annotation using PLINK, genome-wide transposable element and chromatin overlap profiling with BEDTools, and LD-informed enrichment and heritability analyses using LDSC regression.

Data Science Researcher, Department of OB/GYN (PI: Dr. Timothy Dye)

Sep 2024 – Present

- Developed scalable LLM-based frameworks to classify social deprivation levels of 11,000+ unstructured ethnicity self-identifications across 170+ countries from a global COVID-19 survey to support epidemiological & population health analysis, employing fine-tuning & prompt engineering methods.
- Evaluated the performance of several instruction-tuned LLMs against expert-annotated benchmarks, advancing reproducible methods for global health equity research.
- Conducting statistical construct validation analysis comparing AI-derived disadvantage scores to existing healthcare and social variables on multiple global health survey studies.

National Institute of Neurological Disorders & Stroke (NINDS)

Bethesda, MD

Summer Research Intern, Wagner Lab, Neo-Cortex Cerebellum Circuitry Unit

Jun 2024 – Aug 2024

- Studied cerebellar learning mechanisms of long-term dopamine self-stimulation tasks in transgenic mice.

- Developed a computational model for real time video tracking & optogenetic stimulation during open field closed loop conditioned place preference (CPP) experiments using Bonsai visual reactive programming & Python, establishing CPP as an additional paradigm for studying compulsive-reward seeking behavior.
- Applied deep neural networks for pose estimation using DeepLabCut, wrote algorithms in Python to calculate estimated time elapsed for mice in optogenetic reward chambers during CPP trials and visualize motion tracking patterns over the experimental learning period.

National Cancer Institute (NCI)

Summer Research Intern, Vaccine Group, Center for Immuno-Oncology (CIO)

Bethesda, MD

Jun 2023 – Aug 2023

- Investigated the effects of a novel triple combination therapy of a vaccine & two anti-cancer agents in treating HPV-related malignancies on immune protein levels within tumor microenvironment & periphery of pre-clinical murine mice models, as well measuring tumor immune cell populations and anti-tumor activity.
- Performed fluorescence assays on serum & tumor supernatant extracted from TC-1 injected syngeneic mice models, measured gene expression using q-PCR, ran statistical analysis of chemokine/cytokine levels.

LEADERSHIP & ACTIVITIES

University of Rochester Out in STEM (oSTEM)

Co-President (2025-26), Secretary (2024-25)

Rochester, NY

Jan 2024 – Present

- Spearheaded pre-professional events for LGBTQ+ students in STEM fields such as resume workshops, graduate school application sessions, mindfulness discussions and faculty panels, annual Spring Speaker Event with 40+ members in attendance, and organized funding & travel to oSTEM national conference with 8-10 attendees.

University of Rochester Alpha Phi Omega, Mu Lambda

Co-Vice-President of Service (2025-26), Secretary (2024-25)

Rochester, NY

Jan 2024 – Present

- Completed 100+ total hours of community service, participating in weekly service projects through local organizations such as House of Mercy, Rochester Greenovation, Toys for Tots, etc.
- Responsible for managing logistics such as booking rooms for chapter meetings & fellowships and maintaining calendar of all events throughout the semester, organized co-sponsorships with other student organizations.

University of Rochester Neurodiversity Alliance

Co-President (2025-26), Vice-President & Secretary (2023-25)

Rochester, NY

Sep 2023 – Present

- Led a team of 5+ mentors, conducting weekly mentorship sessions for 10+ students with learning disabilities at Rochester Prep Middle School and Joseph C. Wilson Magnet High School, totaling around 60+ hours of service.
- Educate community & share experiences on topics such as neurodiversity, metacognition & self-advocacy through art-based therapy & group activities; plan events and member meetings on campus related to neurodiversity and learning disability awareness.

University of Rochester Office of Residential Life & Housing

First-Year Fellow

Rochester, NY

Sep 2023 – May 2025

- Assist freshmen with navigating first year life, course planning, and academic resources on campus.
- Help plan events such as professor panels, destressing events, and study zone group sessions.

University of Rochester Chemistry Department

Teaching Assistant

Rochester, NY

Jan 2024 – May 2024

- Led weekly workshops and office hours to review material covered in class for *CHEM 132: Chemical Concepts, Systems & Practices II* including quantum mechanics, bond theory, thermodynamics, kinetics, etc.
- Facilitated groups of students with example problems and approaches to understanding & visualizing concepts.
- Assisted with grading midterm exams and extra credit Infographic & Profile a Chemist assignments.

HACKATHON PROJECTS

2025 Biomedical Data Science Hackathon | Numpy, Pandas, LightGBM, Matplotlib, Seaborn

- Developed a machine learning pipeline to predict double gene perturbation expression profiles from single perturbation data, leveraging feature engineering to capture additive and non-linear gene interaction effects.
- Engineered biologically inspired features (average, interaction, difference, etc.) and trained per-gene LightGBM regressors, producing robust genome-wide predictions.
- Achieved an RMSD score of 0.14234, placing first in the entire hackathon out of all competing teams.

2025 RAISE Informatics, & AI Competition | *Pandas, spaCy, Hugging Face, Transformers, BERTopic, Scikit-learn*

- Analyzed 14,013 news headlines to uncover key themes on the promises & challenges of AI across sectors such as education, careers & societal change.
- Performed NLP techniques such as sentiment analysis (Vader, TextBlob, DistilBERT, etc.), N-grams/NER, and topic distribution via BERTopic, and leveraged LLMs via HuggingFace to compare sentiment of key domains and categorize & summarize key topics based on the top words & documents within the dataset.
- Advanced as finalists out of 119 teams, presented findings & insights to industry judges, awarded 2nd Place.

2025 RIT ASA Datafest | *Pandas, NumPy, Seaborn, Matplotlib, Scikit-Learn, Statsmodels*

- Analyzed comprehensive datasets from Savills' commercial real estate firm to derive key insights and identify actionable insights for their advisory services.
- Performed exploratory data analysis & feature engineering, alongside linear regression of different numerical & categorical variables such as space availability, unemployment rate, and total revenue & leasing space.
- Developed prediction model to determine future direction of rent prices in key industries.
- Presented findings to industry judges & awarded 1st Place Gold Medal Award in Best Visualization Category.

2024 Biomedical Data Science Hackathon | *Pandas, NumPy, Seaborn, Matplotlib, XGBoost, Scikit-learn*

- Preprocessed data from 2,999 CD14+ monocyte cells, including 36,601 genes and 108,344 chromatin accessibility peaks. Extracted key features including mean RNA & ATAC counts, and RNA-ATAC correlation. Trained an XGBoost machine learning classifier to predict associations on testing data.
- Achieved an MCC score of 0.9868, placing third overall and first amongst all undergraduate teams.

PRESENTATIONS

National Conference on Undergraduate Research (Poster Presentation) , Richmond, VA	<i>April 2026</i>
7th National Big Data Health Science Conference (Poster Presentation) , Columbia, SC	<i>February 2026</i>
IEEE International Conference on Big Data (Oral Presentation) , Macau, CN	<i>December 2025</i>
URMC Aging Research Day Conference (Poster Presentation) , Rochester, NY	<i>October 2025</i>
Out in STEM National Conference (Poster & Oral Presentation) , Baltimore, MD	<i>October 2025</i>
UR Goergen Institute of Data Science 10th Anniversary Research Fair , Rochester, NY	<i>September 2025</i>
Finger Lakes Science and Technology Showcase (Poster Presentation) , Rochester, NY	<i>April 2025</i>
UR Lightning Talks Competition , Rochester, NY	<i>March 2025</i>
Out in STEM National Conference (Poster Presentation) , Portland, OR	<i>October 2024</i>
NIH Summer Poster Day (Poster Presentation) , Bethesda, MD	<i>August 2024</i>
UR Lightning Talks Competition , Rochester, NY	<i>March 2024</i>
UR Undergraduate Research Poster Expo (Poster Presentation) , Rochester, NY	<i>March 2024</i>
NCI Center for Cancer Research Poster Day (Poster Presentation) , Bethesda, MD	<i>August 2023</i>
NIH Summer Poster Day (Poster Presentation) , Bethesda, MD	<i>August 2023</i>